

PYTHON & DATA SCIENCE

Cocktail Recommender Project Walkthrough

Data collection, vectorization, PCA, clustering, and
similarity-based recommendation, end to end in Python



github.com/swkim94/cocktail



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OVERVIEW

What This Project Demonstrates



API Data Collection

Pulled and parsed 3,081 cocktail records from a public REST API into a clean dataset



Text Vectorization

Converted unstructured ingredient lists into numeric feature vectors for analysis



Dimensionality Reduction

Applied PCA to compress a 416-ingredient feature space while preserving structure



Similarity Search

Built a KNN-based recommender that finds the closest matching cocktail by composition



Custom Visualization

Generated histograms and PCA cluster plots to explain the data, not just model it



Core Python Fluency

pandas, NumPy, scikit-learn, and matplotlib used together on a real, messy dataset



Note: my hands-on instrument-automation experience is LabVIEW-based; this project reflects my Python and data-analysis skills specifically, built independently outside of work.

Import & Structure Raw Data

Source: TheCocktailDB.com public API → parsed into a pandas DataFrame

3,081

Cocktails

416

Unique Ingredients

Top 10 Most Common Ingredients

Lemon Juice

Pineapple Juice

Triple Sec

Water

Gin

Orange Juice

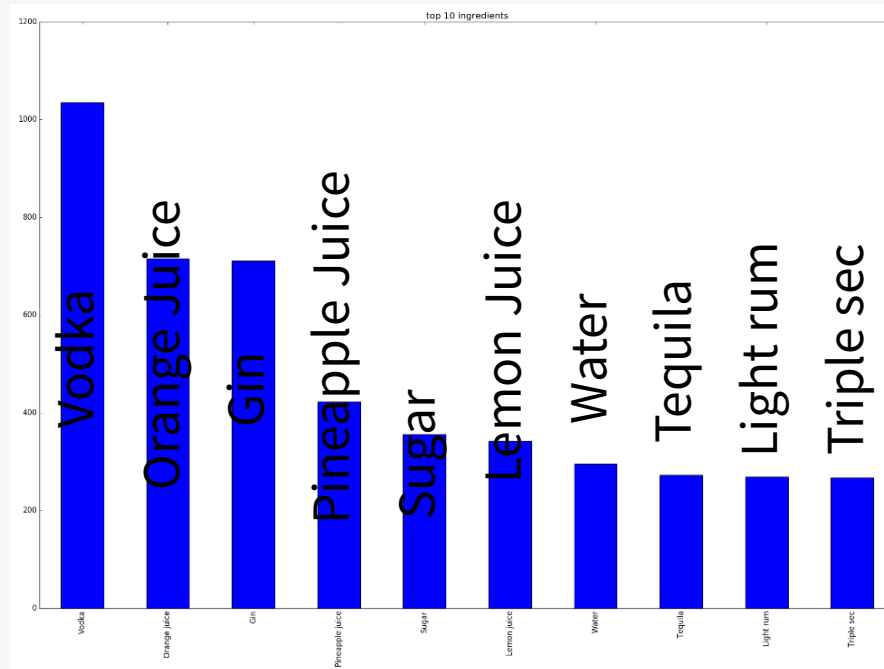
Light Rum

Vodka

Tequila

Sugar

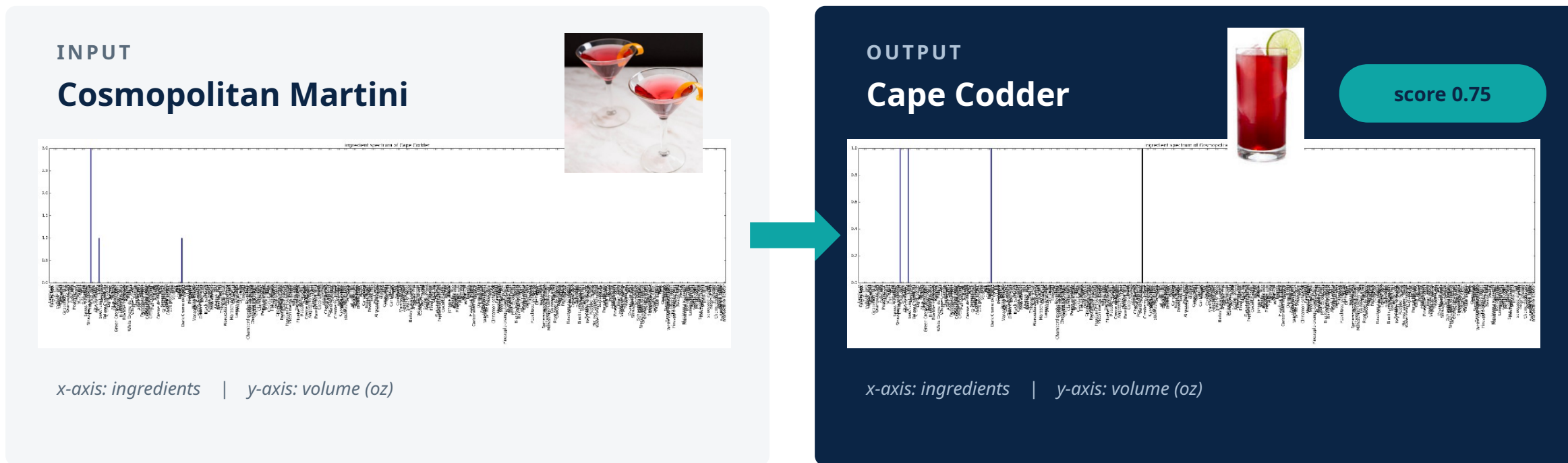
Live pandas DataFrame Output



STEP 2 — RECOMMENDATION ENGINE

Similarity Search (Cosine Similarity)

Each cocktail is represented as a vector of ingredients and volumes. The recommender finds the nearest neighbor in that vector space.



Both charts show ingredient-volume composition — the model matches on the shape of the profile, not just shared ingredient names.

Cosine Similarity algorithm: Normalizing each vector to unit length and then taking the dot product between them.

STEP 3 — UNSUPERVISED LEARNING

PCA & Clustering (KNN)

Cocktails are grouped into 5 clusters using K-Nearest Neighbors (Euclidean distance on raw ingredient-count vectors). PCA then projects the 416-dimension space down to 3 components for visualization.

Vodka

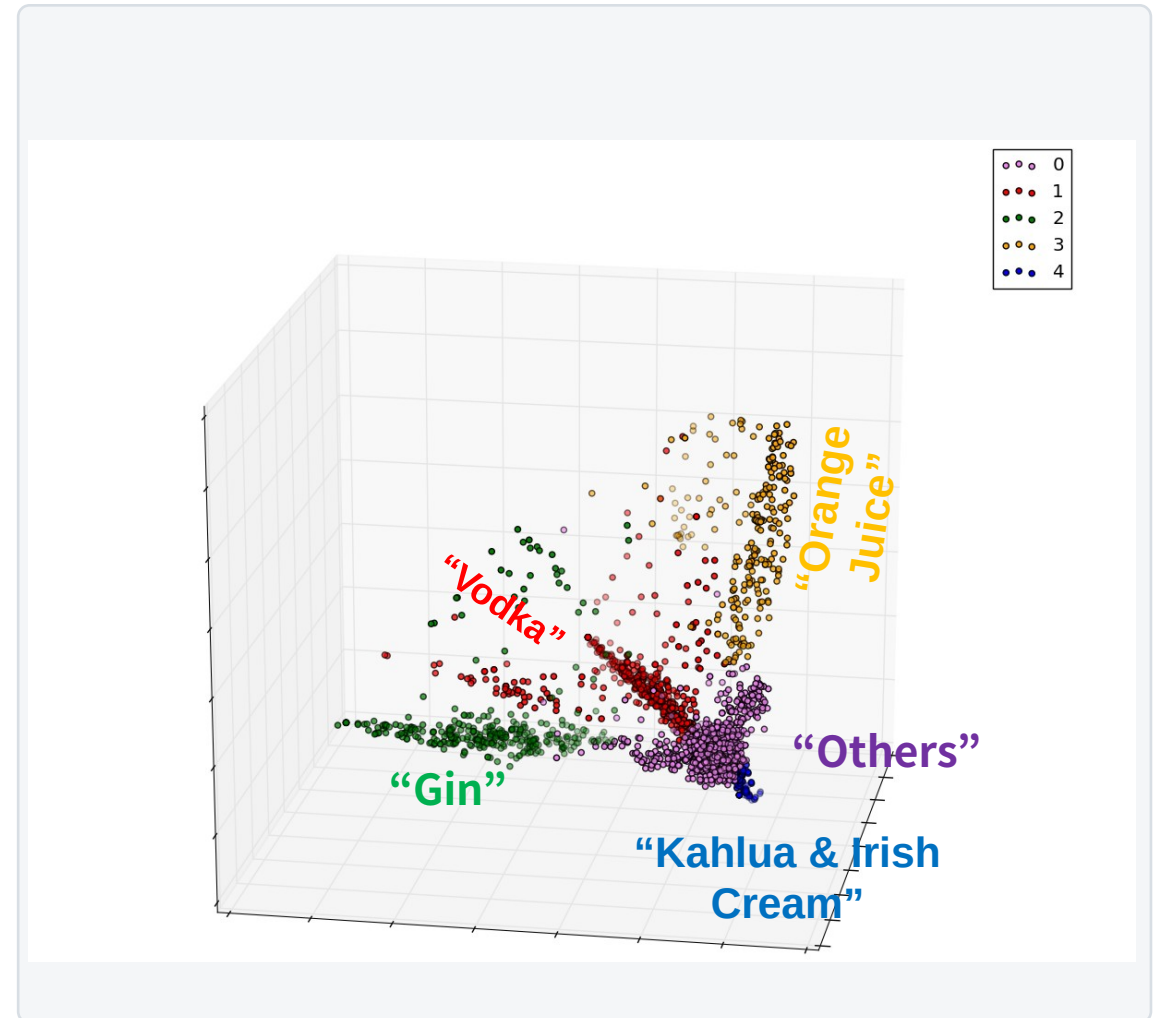
Gin

Orange Juice

Kahlua & Irish Cream

Others

Group labels ('Vodka', 'Gin', etc.) are added afterward based on each cluster's dominant ingredient



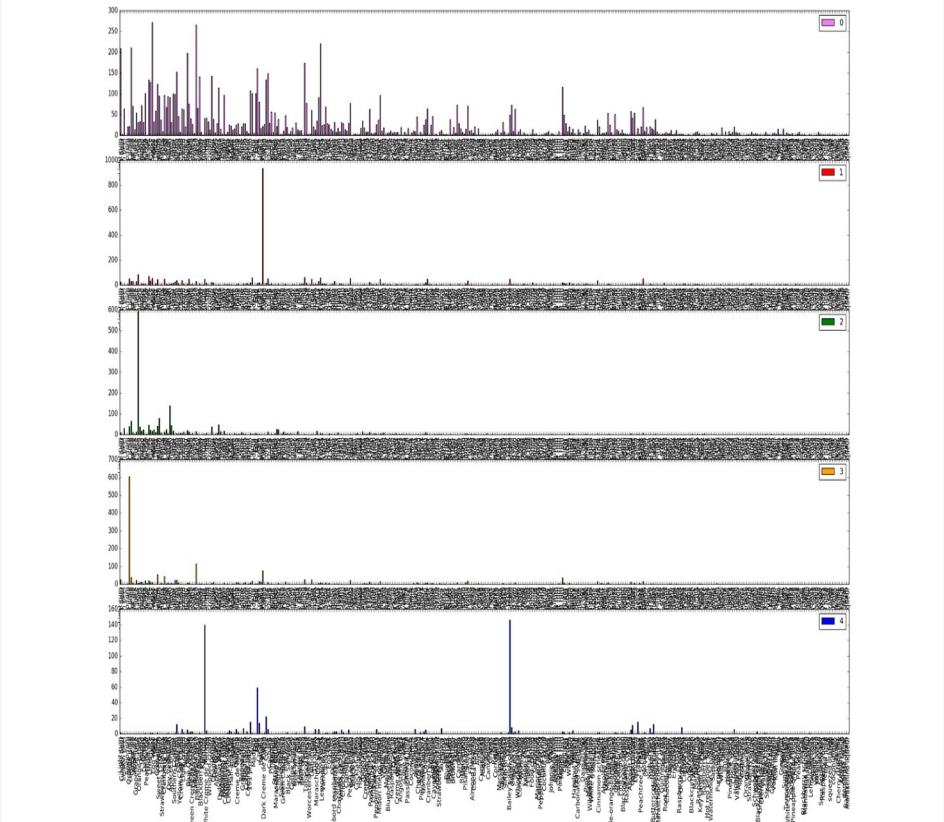
STEP 4 — APPLIED RESULT

Market Segmentation View

Reframing the clusters as segment share turns the same PCA output into a business-facing summary.

Others	58%
Vodka	17%
Gin	12%
Orange Juice	8%
Kahlua & Irish Cream	5%

Each group's profile shown back in the original ingredient space — analogous to reviewing individual univariate test results in semiconductor data.



PCA Is the Same Tool Across Domains



Semiconductor Quality

Tracks multi-variable wafer defects

Screens outliers via multivariate PAT

Pinpoints the equipment fault behind a shift



Smart Factory Automation

Monitors real-time chamber sensor streams

Feeds automated run-to-run control loops

Triggers recipe self-corrections



Chemical Detection (EDX)

Denoises raw spectral data

Maps trace elements clearly

Isolates distinct chemical phases



Market Research

Simplifies complex consumer survey data

Powers 2D brand-positioning maps

Builds a unified customer index score

Thank You

Full source, notebooks, and write-up available on GitHub



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